Session 4
Short/medium/long-term Schedulers

Session 4: Focus

- Schedulers
- Types of Processor Scheduling
  - Long-Term Scheduling
  - Medium-Term Scheduling
  - Short-Term Scheduling

Schedulers

Reference: **Ref1: Chapter 9 on Uniprocessor Scheduling**

Note: Students are expected to read the book (Ref 1) apart from the material given.

What does a Scheduler do?

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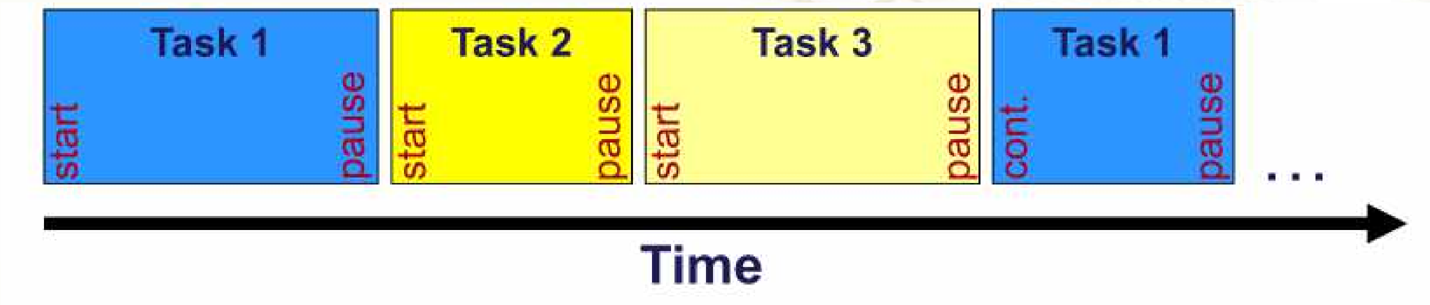
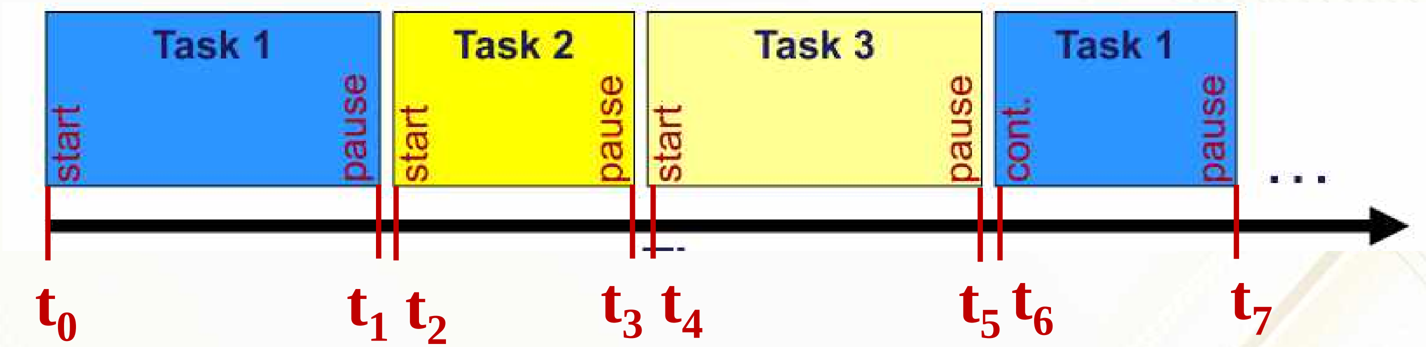


- The main job of the scheduler in the OS is to decide when to stop the currently executing process on the CPU and to choose a next process to run on the CPU.
- When a process is taken out from the CPU, it’s status or context needs to be saved (into the PCB) and load the context of the new process coming into the CPU and allow the new process to run on the CPU.

Quiz 1: When does the Scheduler run?

- If the timings are given as shown below, choose the option which shows all the time slots when the scheduler runs.



A. From $\mathbf{t_0}$ to $\mathbf{t_1}$
B. From $\mathbf{t_0}$ to $\mathbf{t_7}$
C. From $\mathbf{t_1}$ to $\mathbf{t_2}$
D. None of the above

ANS: D

The scheduler gets the control in regular interval or when a currently running process relinquishes the CPU voluntarily and picks the next process to run on the CPU.

Here, the scheduler runs during the time slots: **t_1 to t_2, t_3 to t_4, t_5 to t_6, …**

Scheduling

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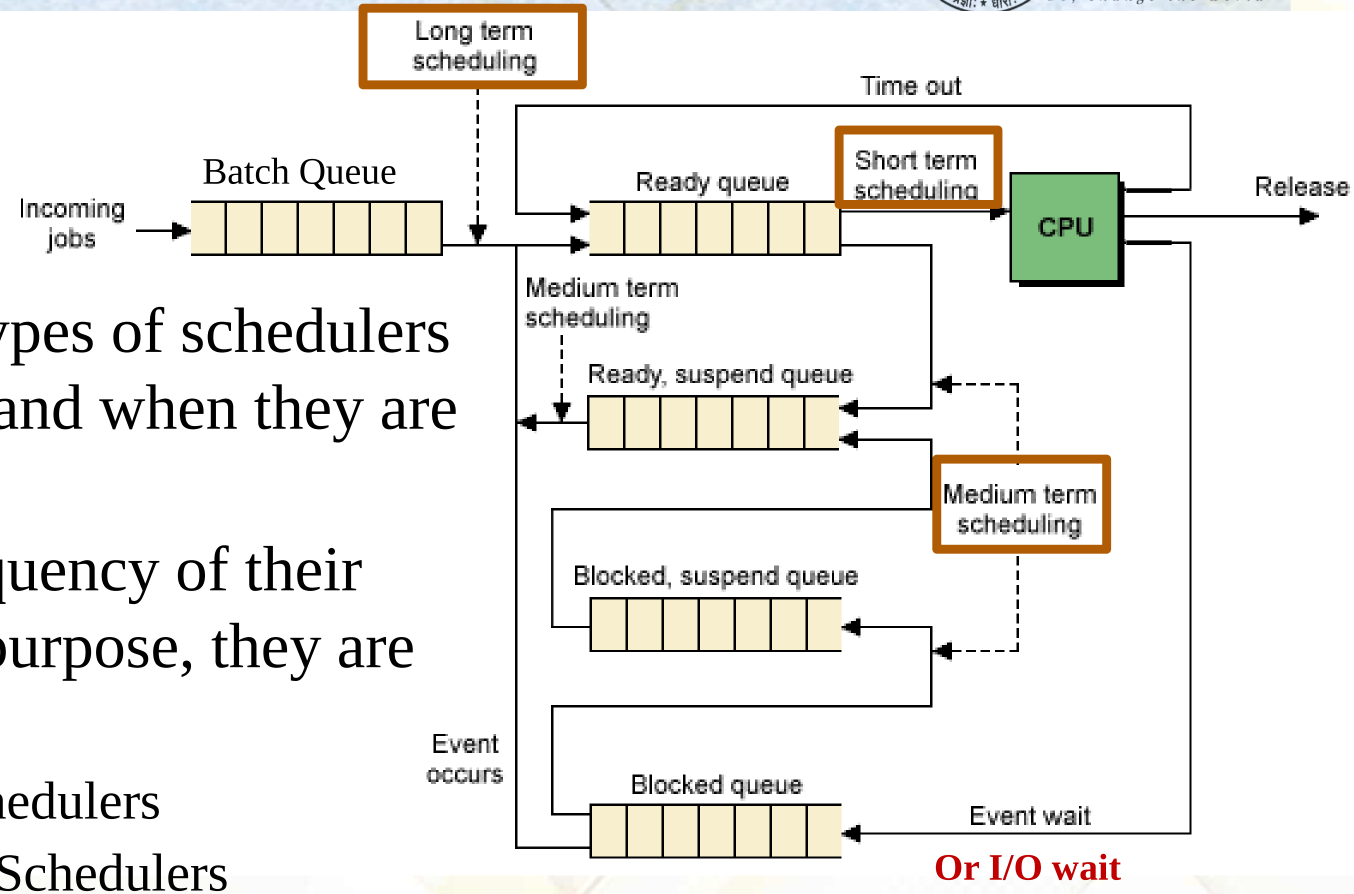
- An operating system must **allocate computer resources** among the potentially **competing requirements** of **multiple processes**.
- In the case of **processor**, the **resource** to be allocated is **execution time** on the processor and the means of allocation is scheduling.
- The **scheduling function** must be designed to satisfy a number of **objectives**, including:
  - Fairness,
  - Lack of starvation of any particular process,
  - Efficient use of processor time, and
  - Low overhead   - Scheduler should do it's job within a minimum time.
- In addition, the scheduling function may need to take into account different levels of priority or real-time deadlines for the start or completion of certain processes.

Types of Schedulers



- There are three types of schedulers based on its role and when they are invoked.
- Based on the frequency of their invocations and purpose, they are categorized as:
  1. Long-Term Schedulers
  2. Medium-Term Schedulers
  3. Short-Term Schedulers
- The picture shows the schedulers and the queues they operate on.

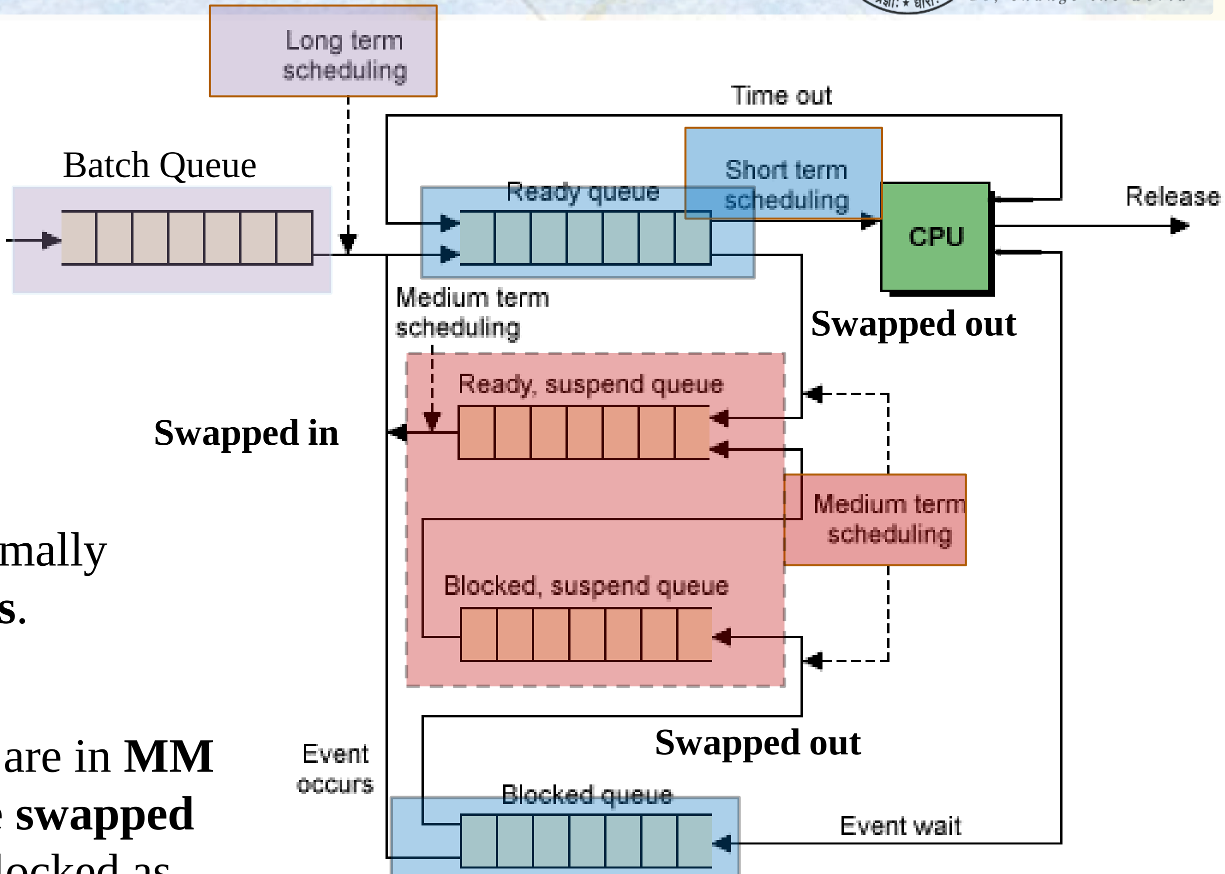
Note: The schedulers being considered here are uniprocessor schedulers where single CPU is present.

Types of Schedulers

Remember only the resident-set of the swapped out processes reside in HDD, whereas the Supended queues are maintained in OS data structures in MM.

Note: **Task** and **Process** are **synonyms**, they could be used interchangeably.

Note: Whereas **Threads** normally mean **light-weight processes**.



In the Main Memory

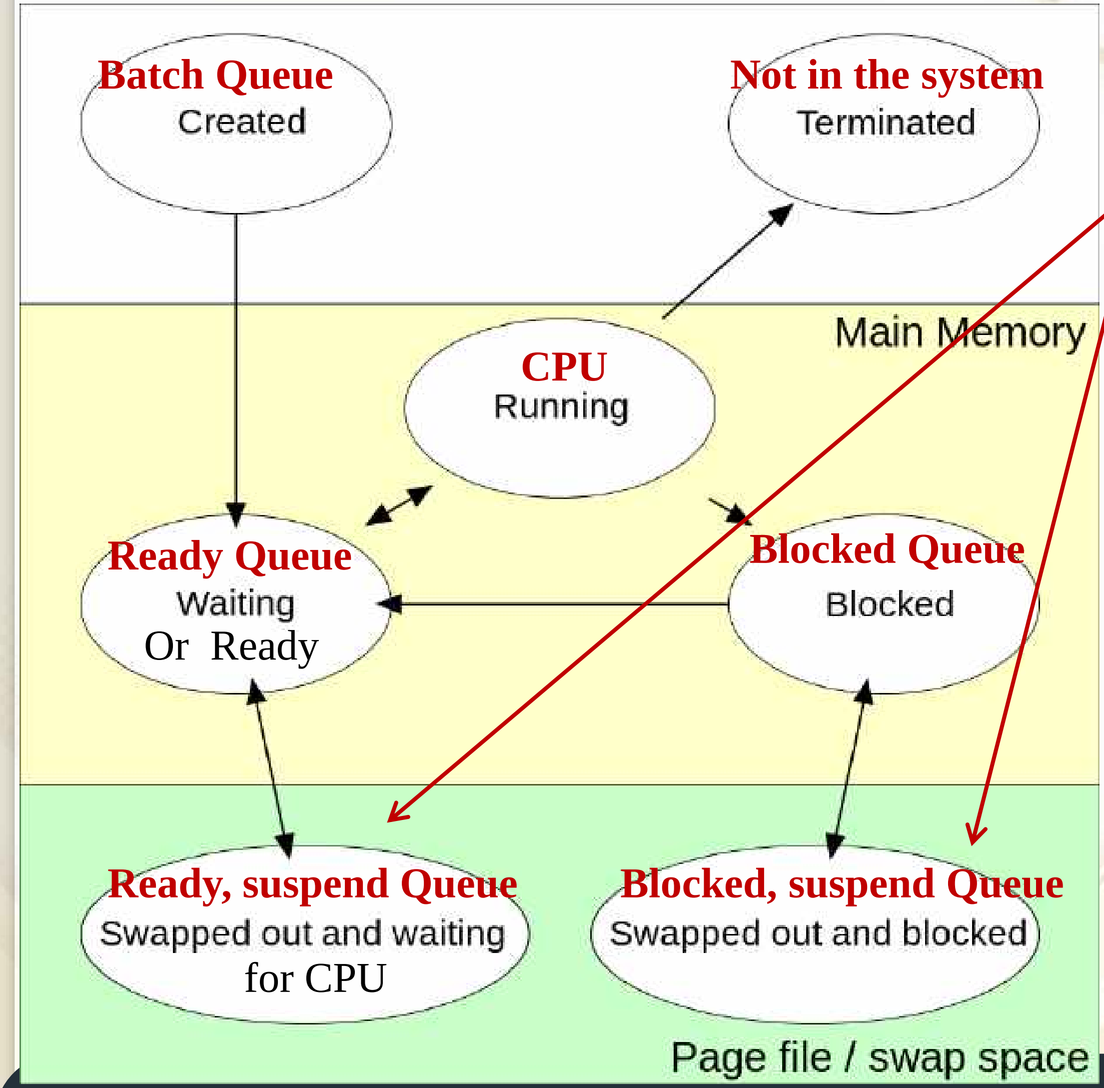
In secondary storage

Definitions:

Blocked: Blocked tasks that are in **MM**

Suspended: Tasks which are **swapped out**. (the tasks could be in Blocked as well as Ready states). Even Ready tasks could be swapped out, and get into Suspend state.

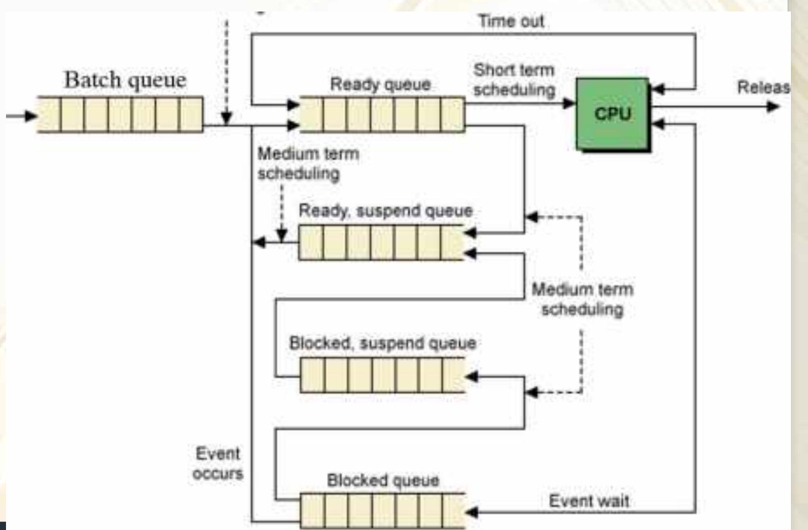
Quiz 2: Tasks in different states and where they reside



In which queue the tasks in each state reside?

Notes:

1. **Swapped out** in **Ready** state.
2. **Swapped out** in **Blocked** state.
3. Medium-term scheduler **swaps in** or **out** processes to **balance** the no. of tasks in the **Main Memory**.



Long-term Schedulers

- If this scheduler is present in the system, it works with batch queue and selects the next batch job to be selected to insert into the Ready queue.
- It admits **new jobs** into Ready queue.
- The primary objective is to have a balanced mix of jobs (Processor and I/O intensive).
- First level of throttle on **resource** utilization.
  - Resource here refers to CPU and MM
- Normally invoked when a completed job departs the system.
- Frequency of invocation is generally **much lower**, which is system and workload dependent.
  - This scheduler can incorporate computationally intensive and complex scheduling algorithms.
  - Because this scheduler is going to be invoked infrequently.

Medium-term Schedulers

- A running process may become blocked by making an I/O request or issuing a system call.
- Since blocked process cannot make any progress until the blocking condition is removed or the event it is waiting for happens
  - May be beneficial to remove it from Main Memory to make room for more number of Ready processes.
  - Medium-term scheduler **Swaps out** processes to Disk.
- When more number of processes become Suspended, no. of processes in the Ready queue depletes, then suspended Ready processes are brought into MM.
- When suspending condition is removed, it **swaps in** the processes from Blocked Suspend queue from HDD.
- This scheduler is invoked when **processes depart** or **no. of processes** in the **Ready queue goes below a specified limit**
- It is present only when **swapping** is supported by the OS.

Note: This scheduler can also swap out **lower priority Ready processes** to HDD when there are many higher priority tasks in the Ready state to be accommodated in MM.

Short-term Schedulers

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- This scheduler allocates the CPU from the pool of processes in the Ready queue which are resident in memory.
- It is invoked for each **process switch** to select the next process to be run.
- It is invoked whenever an event (internal or external) causes the global state of the system to change:
  - On time slice (quantum) completion.
  - Interrupts and I/O completion because of which a blocked process becomes Ready.
  - OS system calls get invoked or get completed.
  - Sending or receiving of signals (we will study about this later).

Session 4: Summary

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References (1 to 5)

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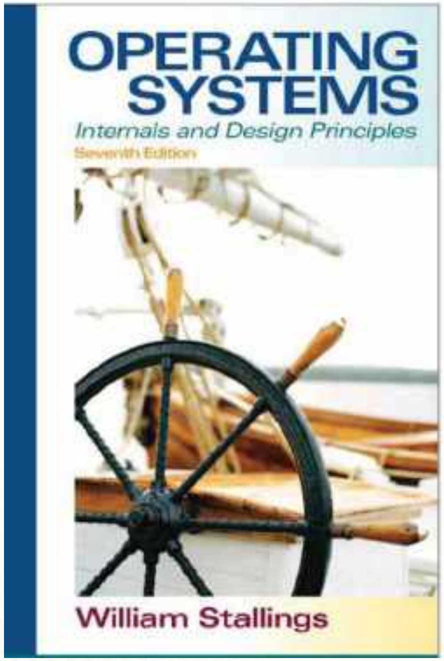
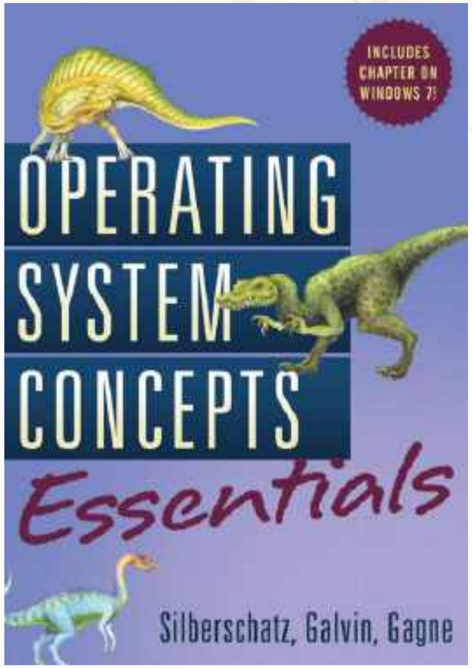
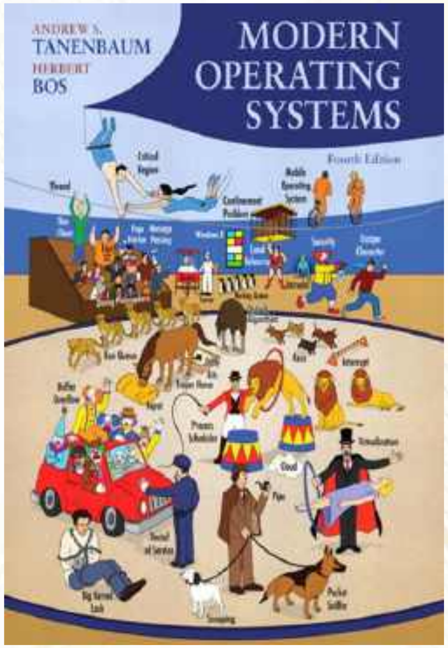
Ref 1



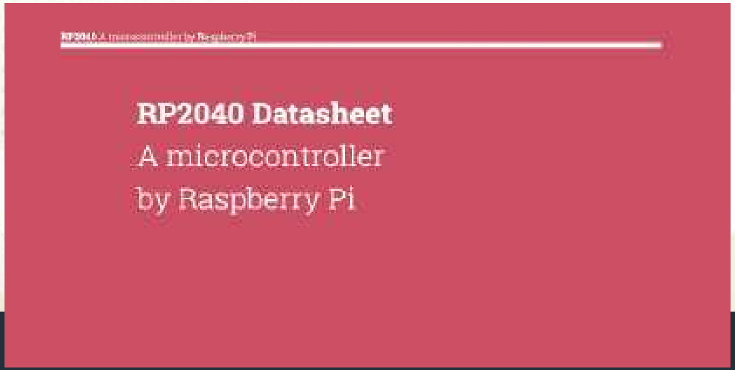
Ref 2



Ref 3



Ref 4



Ref 5